

Hackathon Project Phases Template

Project Title:

Blog Generation Using LLaMA 2 and Streamlit

Team Name:

FAILURES

Team Members:

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Phase-1: Brainstorming & Ideation

Objective:

- **Generate Custom Blog Content:** Use LLaMA 2 and Streamlit to quickly create high-quality, topic-specific blog posts tailored to the user's desired word count and target audience.
- **Streamline Content Creation:** Provide an efficient, user-friendly tool for content creators, marketers, and professionals to produce engaging, contextually relevant blog posts in minimal time.

Key Points:

1. **Problem Statement:** The problem is that content creators, researchers, and professionals often need to produce high-quality written content quickly. This can be time-consuming and sometimes results in writer's block or inconsistency in quality. The goal is to create an AI-powered tool to generate blog content based on user input, streamlining content creation.
 2. **Proposed Solution:** An interactive platform built with Streamlit that uses the LLaMA 2 language model to generate blog posts based on user input. The user specifies a topic, word count, and target audience, and the model generates a relevant and coherent blog post tailored to those inputs.
 3. **Target Users:** Content creators, researchers, marketers, and professionals who need to produce blog posts quickly and efficiently.
 4. **Expected Outcome:** The tool will enable users to generate high-quality blog posts in a matter of minutes, saving time and effort. It will help users overcome writer's block, improve productivity, and create content suited for a specific target audience.
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Phase-2: Requirement Analysis

Objective:

Identify Key User Needs: Understand the requirements of content creators, marketers, and professionals who need quick, tailored blog content based on specific parameters like topic, word count, and audience.

Ensure Seamless Integration: Ensure the LLaMA 2 model and Streamlit interface work together smoothly to deliver fast, accurate, and high-quality blog posts with minimal user effort.

Key Points:

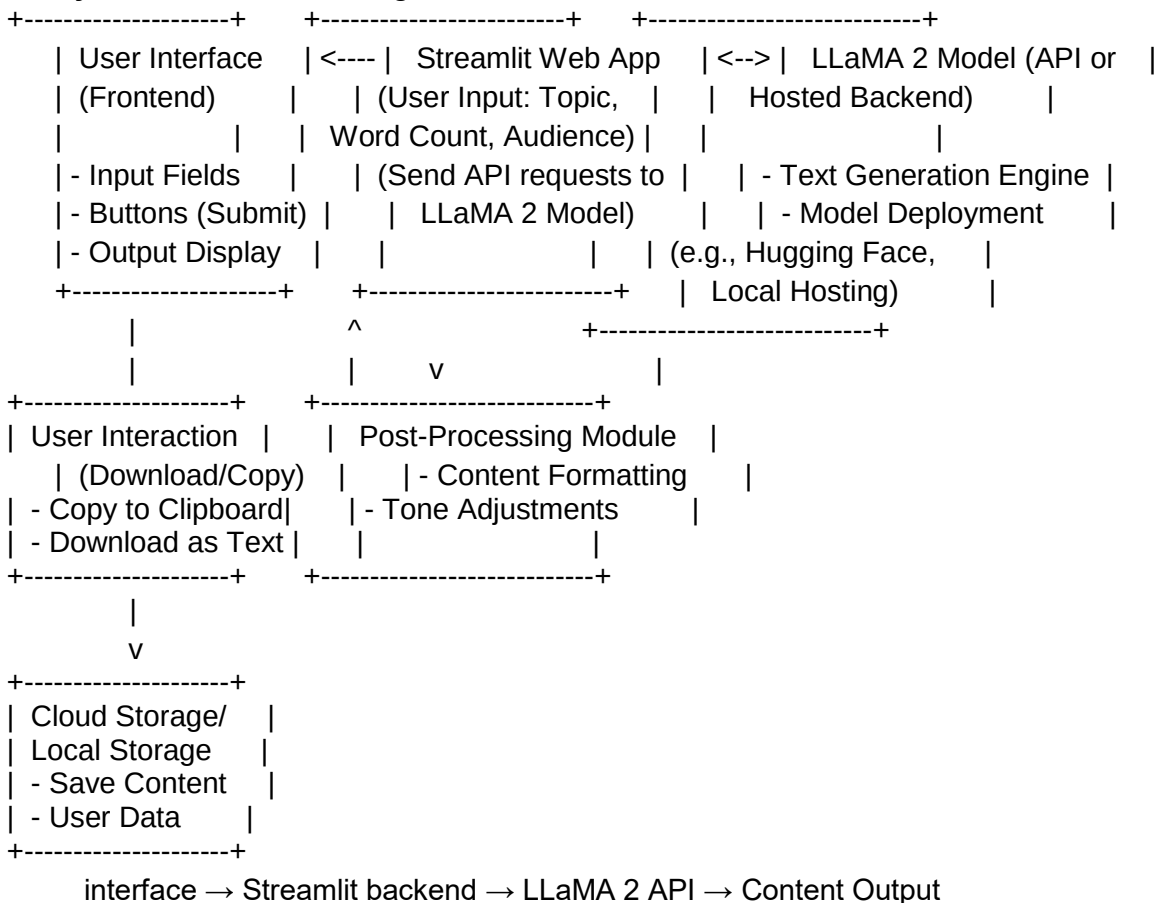
1. **Technical Requirements:** Programming Language: Python
 - Framework: Streamlit
 - AI Model: LLaMA 2 Language Model
 - Hosting/Deployment: Heroku or Streamlit Sharing
 2. **Functional Requirements:** User input fields: Topic, Word Count, Target Audience
 - Content Generation using LLaMA 2
 - Option for users to download the generated blog post
 - Ability to regenerate or edit the content based on feedback
 3. **Constraints & Challenges:**
 - Ensuring the LLaMA 2 model generates coherent and contextually accurate content
 - Fine-tuning the model to handle specific user inputs efficiently
 - Limiting content length and ensuring it meets user specifications for word count
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Phase-3: Project Design

Objective:

Key Points:

1. System Architecture Diagram:



2. User Flow:

- Step 1: User lands on the Streamlit interface
- Step 2: User fills in input fields: Topic, Word Count, Target Audience
- Step 3: Backend calls LLaMA 2 model for content generation
- Step 4: The AI generates the blog post and displays it to the user
- Step 5: User can download the blog post or regenerate if needed

3. UI/UX Considerations:

- Clean, minimal interface with large input fields for topic, word count, and audience
- Easy-to-read output area for the generated blog post
- Option to edit or regenerate content with a simple button click

Phase-4: Project Planning (Agile Methodologies)

Objective:

- Break down the tasks using Agile methodologies.

Key Points:

1. Sprint Planning:

- Sprint 1: Set up Streamlit interface and basic input fields
- Sprint 2: Integrate LLaMA 2 model for content generation
- Sprint 3: Implement content editing and regeneration functionality
- Sprint 4: Testing and deployment

2. Task Allocation:

JATHIN LALITH: Lead backend development (LLaMA 2 integration)

GNS ADITHYA & PAVAN KUMAR: UI/UX design and Streamlit implementation

RAVITEJA: Testing, content evaluation, and deployment

3. Timeline & Milestones: Day 1-2: TO COMPLETE OVERALL PROJECT

Phase-5: Project Development

Objective:

- Code the project and integrate components.

Key Points:

1. Technology Stack Used:

- Programming Languages:* Python
- Frameworks/Tools:* Streamlit, LLaMA 2 API
- Others:* Heroku or Streamlit Sharing for hosting

2. Development Process:

- Developed the user interface using Streamlit
- Integrated LLaMA 2 for text generation based on user inputs
- Ensured content meets specified word count and audience requirements
- Added content regeneration feature for flexibility

3. Challenges & Fixes:

- Challenge: Ensuring LLaMA 2 generates coherent content for niche topics
 - Fix: Fine-tuned the model with additional data and incorporated prompts to guide the generation
- Challenge: Integrating word count limit and maintaining blog quality
 - Fix: Added a mechanism to split content if it exceeds the word limit, without compromising quality

Phase-6: Functional & Performance Testing

Objective:

- Ensure the project works as expected.

Key Points:

1. Test Cases Executed:

- Test 1: User input for general topics (e.g., AI advancements)
- Test 2: User input with a specific audience (e.g., researchers)
- Test 3: User input with maximum word count (500-1000 words)
- Test 4: Test the regeneration feature for content improvement

2. Bug Fixes & Improvements:

- Fixed UI rendering issues on different screen sizes
- Improved content flow to avoid repetition and improve relevance

3. **Final Validation:**

The AI-generated content met the initial requirements and was tailored for specific user inputs (topic, word count, audience).

4. **Deployment (if applicable):**

Deployed the app using Streamlit Sharing. Accessible via [streamlit_app_link].

Final Submission

1. **Project Report Based on the templates**
 2. **Demo Video (3-5 Minutes)**
 3. **GitHub/Code Repository Link**
 4. **Presentation**
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